

36. Quantization Effects in 5G DSP Hardware

Simulate coefficient quantization in a digital beamforming processor and analyze its impact

on beam patterns. (25) (BL5) (WK4)

Given Data:

- **Word Lengths = 8, 10, 12 bits**
- **Number of Antennas = 8**

Tasks:

- 1. Implement beamforming weights.**
- 2. Quantize coefficients.**
- 3. Compare beam patterns.**
- 4. Measure side-lobe levels.**
- 5. Evaluate quantization errors.**
- 6. Recommend a suitable word length.**

1.) Aim:

To simulate coefficient quantization in a digital beamforming processor for a 5G communication system using MATLAB, analyze the effect of different coefficient word lengths (8-bit, 10-bit, and 12-bit) on beam patterns, evaluate quantization errors, measure side-lobe levels, and recommend an appropriate word length for practical DSP hardware implementation.

2. Objectives

- To implement digital beamforming weights for an 8-element antenna array.
- To quantize beamforming coefficients using 8-bit, 10-bit, and 12-bit word lengths.
- To compare the beam patterns obtained before and after coefficient quantization.
- To measure the side-lobe levels for each quantized beam pattern.
- To evaluate the quantization error introduced by finite word-length representation.
- To recommend the most suitable coefficient word length by considering both hardware complexity and beamforming performance.

4. Software Requirements

- MATLAB R2021a or later (or any recent version)

- MATLAB Editor
- Basic MATLAB plotting functions

5. Tools Used

Tool	Purpose
MATLAB	Simulation environment
MATLAB Editor	Writing and executing the program
Plot Function	Plotting beam patterns
Complex Number Operations	Implementing beamforming weights
Quantization Algorithm	Simulating finite word-length representation
Array Signal Processing Concepts	Beam pattern analysis

Code:

```

clc;

clear;

close all;

%% Parameters

N = 8;           % Number of antennas

theta = -90:0.1:90;    % Observation angles

d = 0.5;         % Antenna spacing (lambda/2)

steerAngle = 20;    % Beam steering angle

wordLengths = [8 10 12];

```

```
%% Ideal Beamforming Weights
```

```
n = 0:N-1;
```

```
w = exp(-1j*2*pi*d*n'*sind(steerAngle));
```

```
% Normalize
```

```
w = w/max(abs(w));
```

```
%% Ideal Beam Pattern
```

```
AF_ideal = zeros(size(theta));
```

```
for k=1:length(theta)
```

```
    a = exp(1j*2*pi*d*n'*sind(theta(k)));
```

```
    AF_ideal(k) = abs(w'*a);
```

```
end
```

```
AF_ideal = AF_ideal/max(AF_ideal);
```

```
figure;
```

```
plot(theta,20*log10(AF_ideal),'k','LineWidth',2)
```

```
hold on
```

```
legendText = {'Ideal'};
```

```
fprintf('\n-----\n');  
fprintf('Quantization Analysis\n');  
fprintf('-----\n');
```

```
%% Loop for Different Word Lengths
```

```
for idx=1:length(wordLengths)
```

```
    B = wordLengths(idx);
```

```
    %% Quantization
```

```
    scale = 2^(B-1)-1;
```

```
    wr = round(real(w)*scale)/scale;
```

```
    wi = round(imag(w)*scale)/scale;
```

```
    wq = wr + 1j*wi;
```

```
    %% Beam Pattern
```

```
    AF = zeros(size(theta));
```

```
    for k=1:length(theta)
```

```
        a = exp(1j*2*pi*d*n'*sind(theta(k)));
```

```

    AF(k)=abs(wq'*a);

end

AF = AF/max(AF);

plot(theta,20*log10(AF),'LineWidth',1.5)

legendText{end+1}=sprintf('%d-bit',B);

%% Quantization Error

qError = norm(w-wq);

%% Side Lobe Level

pattern_dB = 20*log10(AF);

[peak,loc]=max(pattern_dB);

temp = pattern_dB;

temp(max(1,loc-20):min(length(temp),loc+20))=-Inf;

SLL=max(temp);

```

```

fprintf('\nWord Length = %d bits\n',B);

fprintf('Quantization Error = %.6f\n',qError);

fprintf('Side Lobe Level = %.2f dB\n',SLL);

end

xlabel('Angle (Degrees)')
ylabel('Normalized Gain (dB)')
title('Beam Pattern Comparison')
grid on
axis([-90 90 -60 5])
legend(legendText)

%% Recommendation

fprintf('\n-----\n');
fprintf('Recommendation\n');
fprintf('-----\n');

fprintf(['8-bit : Lowest hardware cost but highest error.\n']);
fprintf(['10-bit: Good compromise between accuracy and complexity.\n']);
fprintf(['12-bit: Best beam pattern, lowest error, highest hardware cost.\n']);

fprintf('\nRecommended Word Length = 10 bits for most 5G DSP hardware.\n');

```

Output:

Figure 1 ×

